SecureSentinel

An Autonomous AI Agent That Investigates Scams and Guides You to Safety

> **One-line pitch:** SecureSentinel doesn't just warn you about scams — it investigates them, stops them, and gets you to where you actually meant to go.

Theme: Smart Automation (primary) · Cybersecurity (secondary)   **Category:** Software Edition

1. The Problem

Modern phishing and social-engineering attacks have moved far past poorly-written fake emails. Attackers now build convincing banking-login clones, government-service impersonation pages, fake courier/delivery payment requests, fake KYC-update pages, fraudulent investment platforms, and QR-code "urgent verification" scams.

The core failure isn't the technology — it's the human on the other side of the screen. A security tool can say *"Potential phishing site detected"* and a confused, pressured user will still click **"Continue anyway."** Detection alone is not enough: the system needs to understand what the user was trying to do, and safely help them get there — not just block them and walk away.

2. Who This Actually Protects

Design principle: complexity belongs to the machine, clarity belongs to the user.

Elderly / digitally inexperienced users
Get an urgent "your account will be blocked" message and can't tell a fake bank page from a real one. Needs a plain-language answer, not a technical warning.

Students
Scholarship portals, internship forms, and fake placement links exploit urgency and hope ("limited seats," "verify now").

Families
A tech-savvy son or daughter can protect parents without hovering over their shoulder every time they browse.

First-time digital-banking users
A huge population in India moving from cash to UPI/net-banking, with no built-up instinct for spotting a fake page.

This is fundamentally an **accessibility and dignity problem**, not just a security one — the goal is letting people participate safely in digital India without requiring them to become cybersecurity experts first.

3. SDG Alignment

SDG	Contribution
1 — No Poverty	Prevents financial loss that disproportionately hurts low-income, first-time digital users with the least recourse.
3 — Health & Well-Being	Reduces the psychological stress and shame scam victims — especially the elderly — experience after being defrauded.
10 — Reduced Inequalities	Closes the digital-literacy protection gap so less technical users get the same protection as power users.
16 — Strong Institutions	Builds public trust in digital government and banking services by cutting down successful impersonation fraud.

4. The Core Idea — Detect → Investigate → Reason → Correct Path → Recover

This loop is the heart of the project — everything else is an upgrade layered on top of it.

1	**Detection**	ML classifier + heuristics + LLM check flag a suspicious link or message
2	**Investigation**	Agent opens the page in an isolated browser, alongside the real trusted site
3	**Reasoning**	Fuses every signal into one explainable verdict — not a black-box score
4	**Correct Path** (novel feature)	Identifies what the user actually wanted, blocks the fake, opens the real destination
5	**Recovery**	If credentials were already entered, guides the user through safe next steps

Why "Correct Path" is the differentiator

Every existing phishing tool stops at "BLOCKED" — security as *restriction* that leaves the user stranded and annoyed enough to disable the tool. SecureSentinel instead asks: **what was this person actually trying to do?** If someone searched "SBI net banking login" and clicked a sponsored scam result, the system identifies the real intent and opens the legitimate site directly. Security becomes **assistance**, not friction — this is the single feature to lead every pitch and demo with.

Explainable reasoning, not a black box

● **High-Risk Phishing** — Claimed organization: SBI. Domain resembles the real bank domain. Login form detected requesting password + OTP. Urgency language detected ("blocked today," "immediately"). Visual layout closely matches the real SBI login page. **Verdict: 94% likely credential phishing.**

5. What Makes It Trustworthy, Not Just Clever

The investigation agent never touches the user's real browser session. It operates in a fully isolated context with no cookies, no saved passwords, and no personal data — so an autonomous agent can safely click links, fill test forms, and inspect suspicious pages without ever putting the user at risk. A separate trusted-reference context is used only to verify the real organization's official site, keeping the two investigations from ever contaminating each other.

Autonomy Level	Examples
Automatic	Analyze URLs, inspect suspicious pages in isolation, block confirmed-malicious pages, open trusted reference pages
Requires user confirmation	Password-reset guidance, changing security settings, sending family notifications
Never done autonomously	Financial transfers, purchases, irreversible account actions, submitting real credentials anywhere

6. System Architecture

User encounters a link / message

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Chrome Extension

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Threat Detector — LightGBM + heuristics + LLM

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Investigation Browser (isolated, suspicious page)	Trusted Reference Browser (real organization site)	Visual / Logo Comparison (screenshot similarity)

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Evidence Fusion — stacked meta-model

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Explainable Verdict + Response Planner

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Allow	**Correct Path** redirect to real site	**Block**

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Human approval for any consequential action → Recovery / Notification

7. Build Roadmap

Core build (demoed live)

- Detection engine (retrained classifier + heuristics + LLM check)
- Investigation agent (isolated vs. trusted browser contexts)
- Evidence fusion + explainable reasoning
- **Correct Path** redirection — the headline feature
- Guided recovery (instructions only, no live account actions)
- One flagship demo: bank/KYC scam, end-to-end

Upgrades if time allows

- Visual / logo similarity detection (embedding cosine similarity)
- Threat-intel feed integration (PhishTank, OpenPhish)
- Opt-in family-protection notifications
- Multi-agent decomposition once the core pipeline is solid
- Simple SOC-style incident dashboard for enterprises

8. Success Metrics

Security
Detection precision, recall, false-positive rate, impersonation-detection accuracy.

Agent performance
Investigation completion time, correct-path redirection accuracy, autonomous task success rate.

Human impact
Can a non-technical user understand the verdict unaided? Time saved per incident.

Safety
Rate of unsafe autonomous action, human-approval rate, false-intervention rate.

Problem statement in one line: Develop an autonomous AI agent that detects, investigates, and responds to phishing and social-engineering attacks in real time — safely operating an isolated browser to verify claimed identities against trusted sources, and rather than simply blocking the user, identifying their original intent and redirecting them to the legitimate destination, with human-in-the-loop control over any high-risk action and a fully explainable, auditable decision trail.